



$$\vec{OC} = m \, \boldsymbol{v} \, ,$$

$$\vec{AO} = \frac{m_1(\boldsymbol{p}_1 + \boldsymbol{p}_2)}{m_1 + m_2} \, ,$$

$$\vec{OB} = \frac{m_2(\boldsymbol{p}_1 + \boldsymbol{p}_2)}{m_1 + m_2} \, .$$